Project Phoenix - Integrated Adaptive Powertrain

System Whitepaper v1.0

Project Phoenix

Document status: Phase 1 simulation complete - no hardware gate cleared

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Classification: External distribution permitted with simulation disclaimer

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Disclaimer: This document reports simulation-validated engineering projections. No road-test, EPA/WLTP certification, or homologation claims are made. Prefix all external statements with ?simulation shows...? Measured bench prototypes are the Seed-phase deliverable.

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1. Executive Summary

Project Phoenix is an integrated series-hybrid powertrain combining:

1. ATPE - crankless three-tier free-piston linear generator
2. PCMRITMS - phase-coordinated multi-rotor inertial torque buffer
3. 20 kWh LFP battery - low-frequency energy store and urban EV range
4. Unified controller - nested slow/fast loops that keep the engine off transients

Physics-based computer simulation stress-tested this architecture across **thirteen validation studies**, six vehicle bodies, temperatures from -10 degC to +40 degC, regulatory-style drive cycles, full payload, cold starts, plug-in charging, 250,000 km ageing, comparison against a conventional 2.0 L turbo, and single-cartridge fault tolerance.

Core conclusion (simulation): a small, cool-running, lifetime-lasting battery plus an inertial sprint reserve plus an efficient generator can deliver strong performance with competitive lifetime cost and carbon - if hardware matches the model assumptions.

What has not been done: building and measuring real cartridges, rotors, or a mule vehicle.

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2. System Architecture

2.1 Energy flow

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PETROL
  ?
  ?
ATPE (3-tier free-piston ring)     ? slow loop: 10-100 ms
  ?
  ?
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DC bus (400-800 V)
  ?
  ??? PCMRITMS (inertial buffer)  ? fast loop: 1-5 ms
  ??? LFP battery (20 kWh)
  ?
  ?
Traction motor(s) -> Wheels
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2.2 Complementary roles

ATPE weakness	PCMRITMS remedy
Tier switching gap (~200-400 ms)	Buffer covers seamlessly
Irregular free-piston ripple	Rotor smoothing
Slow generation setpoint	Millisecond inertial response

PCMRITMS weakness	ATPE remedy
118 kJ depletes in ~1.3 s at full discharge	Engine refills during sustained load
Needs primary energy source	Efficient continuous generation
Round-trip losses	ATPE efficiency advantage offsets

2.3 Reference sizing - AWD SUV (Phase 1)

Subsystem	Parameter	Value
ATPE stack	Peak electrical	230 kW (8 cartridges, 4/2/2 mix)
ATPE stack	Continuous	~100 kW
PCMRITMS	Stored energy	118 kJ
PCMRITMS	Continuous discharge	90 kW
PCMRITMS	Brief burst	140 kW (~1.3 s energy-limited)
PCMRITMS	Peak torque boost	+34.9% (242.8 vs 180 N·m)
Battery	Capacity	20 kWh LFP
Battery	Max discharge (default)	120 kW
Battery	Right-sized discharge (SUV)	90 kW
Battery	Max charge	80 kW
Battery	Charge-sustaining SoC	55%
Traction	Peak power	222 kW
Traction	Peak torque	>=350 N·m
Vehicle	0-100 km/h target	<8 s
Vehicle	Top speed	180 km/h

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3. Unified Control

Three nested loops with strict authority:

Loop A - Mode (100-500 ms): electric-only vs charge-sustaining from filtered demand and SoC floor (25%).

Loop B - ATPE setpoint (10-100 ms): five-second low-pass filtered load plus SoC correction -> smallest sufficient tier set.

Loop C - Buffer arbitration (1-5 ms):

1. PCMRITMS supplies or absorbs mismatch first.
2. Battery covers remainder.
3. Any residual is recorded as capability shortfall - energy is never silently invented.

Surplus generation refills PCMRITMS first, then charges the battery.

Power balance (every timestep): generator + battery + buffer = demand + losses.

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4. Validation Studies - Results (simulation)

Study	Question	Headline result
A - Learned controller	Can control learn from data?	Matches rule-based with fair charge-sustaining correction
B - Battery thermal	Heat and ageing?	+1 degC in normal driving; 0 replacements over vehicle life
C - Sensitivity	Which inputs matter?	Aerodynamics > weight at highway; no single magic factor
D - Economics	True cost and CO??	Battery embodied CO? = 4.7% of SUV lifecycle
E - Smart flywheel timing	Cleverer sprint use?	No benefit - energy-limited, not power-limited
F - Right-sizing	Oversized parts?	SUV battery can drop 120 -> 90 kW discharge
G - Uncertainty	Error bars?	5th-95th percentile bands on all headlines
H - Climate	-10 degC to +40 degC?	12-23% seasonal fuel swing; no thermal derate
I - Regulatory cycles	WLTP/EPA reconstructions?	Rankings match custom cycles
J - Cold start	Cold morning?	Hurts long trips only; urban stays electric
K - Payload	Full load?	More fuel; hill climb still passes
L - Plug-in	Grid charging?	CO? win only on clean grid
M - Ageing	250,000 km?	~23% EV range loss; modest fuel creep
N - ICE benchmark	vs 2.0 L turbo?	28% less fuel (AWD SUV, mixed)

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5. Key Performance Data (simulation)

5.1 AWD SUV headlines

Metric	Value
Highway fuel (charge-sustaining)	4.46 L/100 km
Mixed-cycle fuel	2.35 L/100 km
vs 2.0 L turbo (mixed)	-28% fuel
Mixed-cycle CO?	54 g/km (ICE: 75 g/km)
Lifecycle CO? (cradle-to-grave)	102 g/km
Running cost	EUR 0.072/km [EUR 0.066-0.091]
Battery replacements (250k km)	0
WLTP-class fuel	4.61 L/100 km (ICE: 7.30)
Seasonal fuel swing	11.6%
Cold-start fuel penalty	+19.3%
Full-payload fuel penalty	+19.2%

PHEV CO? (clean grid)	12 g/km
Fuel drift after 250k km	+44.4%
EV range loss after 250k km	-23.3%

5.2 Fleet - mixed cycle (simulation)

Body	Fuel (L/100 km)	Cost (EUR /km)	CO? (g/km)	WLTP (L/100 km)	Season swing
Hatchback	0.55	0.039	44	3.60	22.7%
Sedan	0.64	0.040	47	3.91	22.3%
Crossover	1.11	0.048	60	4.87	20.2%
AWD SUV	2.61	0.072	102	6.07	11.6%
Van / MPV	2.95	0.077	112	6.59	16.6%
Pickup	3.86	0.092	137	7.83	14.8%

5.3 ATPE vs conventional 2.0 L turbo - full comparison (L/100 km)

Body	Urban	Highway	Tow+grade	Mixed	WLTP-class
AWD SUV	0.00 / 0.00	4.46 / 7.67	6.91 / 11.92	2.35 / 3.26	4.61 / 7.30
Sedan	0.00 / 0.00	0.83 / 1.42	4.64 / 9.21	0.40 / 0.64	1.64 / 2.70
Hatchback	0.00 / 0.00	0.69 / 1.19	4.09 / 8.47	0.34 / 0.56	1.18 / 1.92
Crossover	0.00 / 0.00	1.50 / 2.43	5.58 / 10.37	1.01 / 1.46	3.03 / 4.91
Pickup	0.00 / 0.00	6.17 / 9.68	8.58 / 13.62	4.23 / 7.01	6.81 / 10.75
Van / MPV	0.00 / 0.00	5.01 / 8.29	7.39 / 12.42	2.77 / 5.12	5.38 / 8.55

Format: ATPE / conventional ICE. Urban figures are near-zero when battery is charged.

5.4 Fault tolerance

18 of 18 scenarios PASS - one cartridge offline in any tier, all six bodies pass 9/9 performance targets (acceleration, top speed, 20% gradeability, efficiency).

5.5 Virtual single-cartridge bench

48-cell matrix (4 speeds x 4 loads x 3 tiers): 100% pass rate (simulation).

6. Development Gate Status

Gate	Milestone	Simulation	Hardware
1	Single cartridge bench	48-cell matrix complete	Not started
2	Stable free-piston	Timing model	Not started
3	Linear generator	Parametric model	Not started
4	Multi-cartridge ring	X4-X16 layout study	Not started
5	Vehicle integration	Complete (6 bodies)	Not started
6	AI optimisation	Study complete	Not started
7	CAD / packaging	Concept renders	Not started

Software position: vehicle integration gate complete.

Hardware position: no gate cleared.

7. Conclusions (simulation)

Supported by modelling:

- Architecture is internally energy-consistent.
- Small LFP pack runs cool, lasts the vehicle life, low embodied carbon (4.7% of lifecycle).
- Capability comes from stored energy and battery power, not peak kW alone.
- Default battery is over-specced - right-sizing saves cost and weight.
- Smart flywheel timing was tested and rejected - honest negative result.
- One cylinder can fail; all bodies still pass performance targets.

Not validated externally:

- Combustion efficiency on a real rig
- Rotor round-trip efficiency and burst containment
- NVH, packaging, homologation
- Official regulatory test traces (reconstructions used)

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8. What This Document Does Not Claim

Limit	Implication
No physical prototypes	Efficiency, noise, weight are modelled
Combustion is simplified in default model	Detailed physics used only for bench matrix
WLTP/EPA are reconstructions	Not official copyrighted traces
AI controller is experimental	Not ASIL-certified production software
PCMRITMS 75-85% round-trip	Optimistic until bench measured
Patents not yet filed	File before wide public disclosure

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9. Roadmap to Measured Credibility

Step	Months	Deliverable	Model falsified if...
Single ATPE cartridge rig	0-12	Pressure, stroke, kW data	BTE ? 40% at sweet spot
Single PCMRITMS rotor bench	6-18	Torque pulse, round-trip ?	? ? 70% or no +35% boost
Integrated electrical bench	18-24	Shared 800 V bus	Bus instability under spike
Mule vehicle	30-42	Dyno + on-road fuel	Highway fuel ? 6 L/100 km
Emissions + durability	42-54	Euro 7 / 300k km equiv	Regulatory or wear failure

Seed funding (illustrative): $8-15M over 18 months - two bench prototypes with measured data.

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10. One-Paragraph Summary

Project Phoenix pairs a crankless three-tier linear generator with a phase-coordinated inertial torque buffer and a small lifetime battery on a unified 800 V DC bus. Thirteen simulation studies across six vehicle types support the conclusion that this architecture can match strong performance with approximately 28% less fuel than a conventional 2.0 L turbo on the same SUV, zero battery replacements over the vehicle life, and **fault tolerance with one cylinder offline**. Simulation shows these outcomes; measured prototypes are the required next step.

Companion documents: ATPE Whitepaper v1.0; PCMRITMS Whitepaper v1.0.